

ROOT CAUSE ANALYSIS REPORT

Incident Title — Service / Component Affected

Incident ID	INC-YYYY-XXXX
Severity / Priority	P1 — Critical
Detection Date/Time	YYYY-MM-DD HH:MM UTC
Resolution Date/Time	YYYY-MM-DD HH:MM UTC
Total Duration	X hours Y minutes
Incident Commander	Name / Handle
RCA Author	Name / Handle
Document Version	v1.0 — Draft
Review Status	Pending Approval

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1. Executive Summary

[2–4 sentences. Describe what broke, the customer impact, the root cause in plain language, and what prevents recurrence. Written for non-technical stakeholders.]

Incident summary	[One sentence: what failed and what the user experienced]
Root cause (brief)	[Plain-language cause — no jargon]
Business impact	[Revenue, SLA, reputation]
Resolution	[What was done to fix it]
Prevention	[What prevents recurrence]

2. Impact Assessment

Category	Details	Metric / Value
Users affected	[Count or percentage of affected user base]	[e.g., 12,400 users (~8%)]
Services affected	[List impacted services / endpoints]	[e.g., auth-svc, api-gateway]
Error rate delta	[Normal vs peak error rate]	[e.g., 0.1% → 47%]
Latency impact	[p99 latency change]	[e.g., 200ms → 8,400ms]
Revenue / SLA	[Financial or SLA breach details]	[e.g., \$X, SLA breached]
Regions affected	[Cloud regions / data centres]	[e.g., ap-south-1]
Customer complaints	[Support ticket or social media volume]	[e.g., 34 tickets]

3. Incident Timeline

All timestamps in UTC. Include every detection, escalation, mitigation, and resolution event.

Timestamp (UTC)	Actor	Event / Action	Category
YYYY-MM-DD HH:MM	[System/Person]	[First anomaly observed — describe what was seen]	First anomaly
YYYY-MM-DD HH:MM	[Monitoring]	[Alert fired — describe which alert and threshold]	Alert
YYYY-MM-DD HH:MM	[On-call SRE]	[Incident declared, war room opened]	Escalation

YYYY-MM-DD HH:MM	[On-call SRE]	[Initial investigation started, logs examined]	Investigation
YYYY-MM-DD HH:MM	[On-call SRE]	[Root cause hypothesis formed]	Diagnosis
YYYY-MM-DD HH:MM	[Eng Lead]	[Containment action taken — describe action]	Containment
YYYY-MM-DD HH:MM	[On-call SRE]	[Fix deployed / rollback executed]	Resolution
YYYY-MM-DD HH:MM	[Eng Lead]	[Services confirmed healthy, all-clear issued]	All-clear

4. Root Cause Analysis

4.1 Root cause statement

[State the single verified root cause clearly. Example: A misconfigured connection-pool max-size (set to 10 instead of 100 during the config migration) caused all auth-service threads to block waiting for DB connections under normal load, resulting in cascading timeouts.]

4.2 5-Whys analysis

Why #	Question	Answer
1	[Why did X happen?]	[Because of Y — provide specific, factual answer]
2	[Why did X happen?]	[Because of Y — provide specific, factual answer]
3	[Why did X happen?]	[Because of Y — provide specific, factual answer]
4	[Why did X happen?]	[Because of Y — provide specific, factual answer]
5	[Why did X happen?]	[Because of Y — provide specific, factual answer]

4.3 Fishbone (Ishikawa) summary

Category	Contributing cause identified
People	[Describe the causal factor in this category, or 'N/A']
Process	[Describe the causal factor in this category, or 'N/A']
Technology	[Describe the causal factor in this category, or 'N/A']
Environment	[Describe the causal factor in this category, or 'N/A']
Measurement	[Describe the causal factor in this category, or 'N/A']
Management	[Describe the causal factor in this category, or 'N/A']

5. Contributing Factors

Secondary causes that enabled the incident to occur or prolonged its impact.

Factor	Category	Description	Severity
Monitoring gap	Technology	[No alert on pool utilisation > 80%]	High
Config validation	Process	[Migration lacked automated value validation]	High
Missing runbook	Process	[No documented procedure for pool exhaustion]	Medium
On-call knowledge	People	[On-call had no prior exposure to this failure mode]	Low

6. Detection & Response

How detected	[Alert / monitoring / customer report / manual observation]
Time to detect (TTD)	[Time from first anomaly to alert firing]
Time to acknowledge (TTA)	[Time from alert to on-call engagement]
Time to mitigate (TTM)	[Time from engagement to service restored]
Total incident duration	[First anomaly to all-clear]
Initial response action	[First action taken by responder]
Escalation path	[Who was paged, in what order, and when]
Communications sent	[Status page update, Slack announcement, customer email — yes/no]
Monitoring gaps identified	[What alerts were absent that would have helped]

7. Corrective & Preventive Actions (CAPAs)

Each action must have a single named owner, a tracked ticket, and a firm due date.

Action description	Type	Owner	Due date	Ticket	Status
Add circuit breaker for DB connection pool	Fix	@owner	YYYY-MM-DD	ENG-XXXX	Open

Add p99 latency alert for auth service	Monitor	@owner	YYYY-MM-DD	ENG-XXXX	In progress
Implement config schema validation in CI pipeline	Process	@owner	YYYY-MM-DD	ENG-XXXX	Open
Write runbook for DB pool exhaustion scenario	Process	@owner	YYYY-MM-DD	ENG-XXXX	Open
Add integration test for connection pool config values	Test	@owner	YYYY-MM-DD	ENG-XXXX	Open
Conduct blameless post-mortem training for on-call team	Training	@owner	YYYY-MM-DD	ENG-XXXX	Planned

8. Lessons Learned

8.1 What went well

- [Alert fired within target SLA]
- [On-call identified root cause quickly]
- [Rollback completed within X minutes]
- [Communication to stakeholders was timely]

8.2 What could be improved

- [No automated validation for config values]
- [Pool utilisation had no monitoring pre-incident]
- [Responder had no runbook — worked from scratch]
- [Customer comms delayed by X minutes]

8.3 What surprised the team

- [Describe any unexpected findings or failure modes]
- [Note any assumptions that were proven wrong]

8.4 Knowledge to share with other teams

- [Cross-team learning point 1]
- [Cross-team learning point 2]

9. Approvals & Sign-off

All reviewers must approve before this document is marked Closed.

Reviewer name	Role / Team	Date reviewed	Signature	Status
[Name]	Engineering Manager	[Date]	[Signature]	Pending
[Name]	SRE Lead	[Date]	[Signature]	Pending
[Name]	Product Owner	[Date]	[Signature]	Pending
[Name]	CISO / Security (P1 only)	[Date]	[Signature]	Pending

Document revision history

Version	Author	Date	Changes
v1.0	[Author name]	[YYYY-MM-DD]	Initial draft